



What Makes a Level Hard?

We tried four ways to make behavior predict difficulty. Counting the pieces beat them all.

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Project Repo

The Problem Statement

Predict a level's human win rate from the level alone — no pre-launch playtesting.

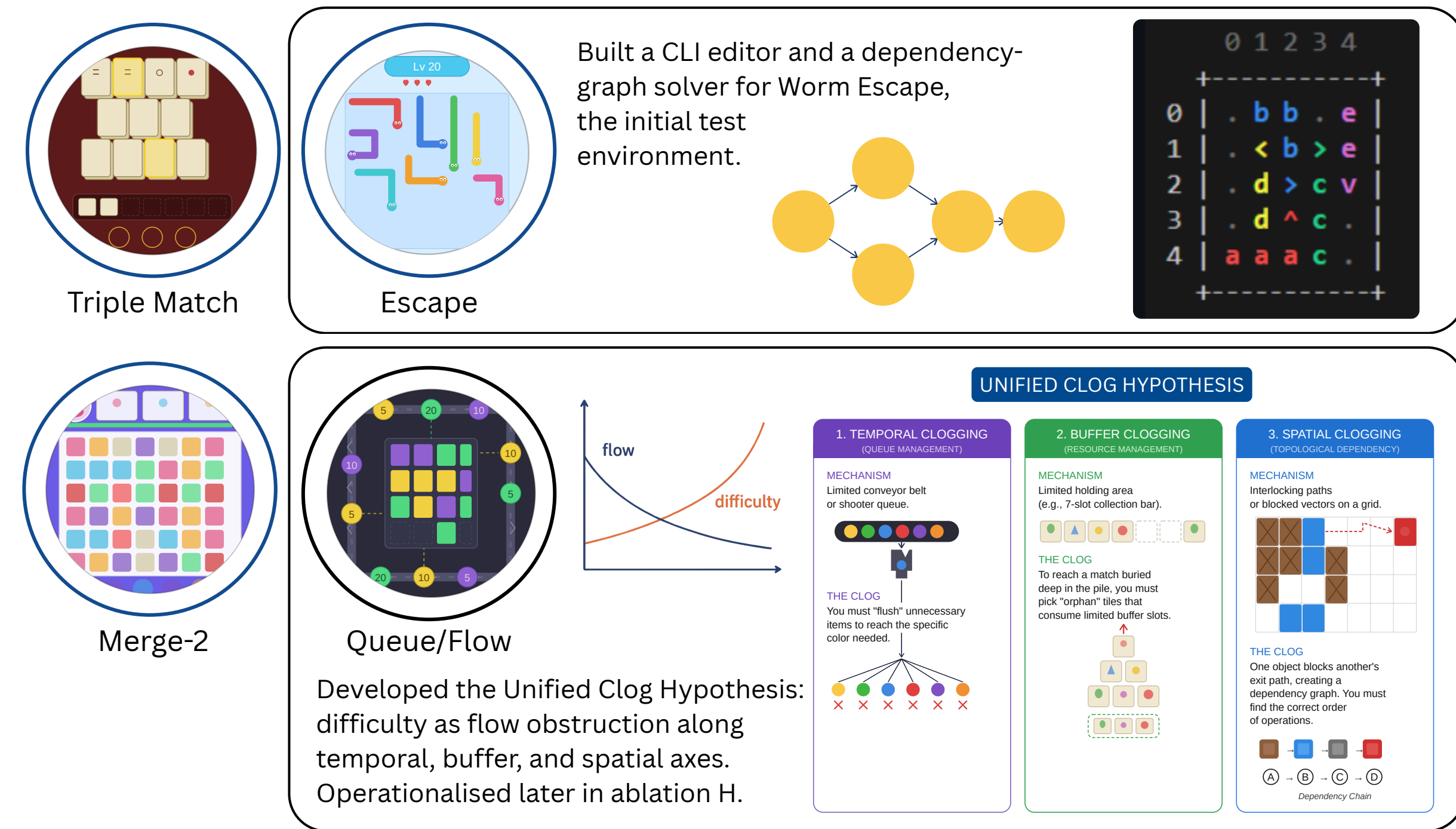
Why it matters: Procedural generation needs difficulty estimates before players see the level

Input: 99 real Hex Fall levels

Target: Real human win rate per level (from Paxie)

Early Phase

The project's first two months focused on a broader question: where in Paxie's portfolio is difficulty most tractable, and what makes a level hard in the first place?



The scope narrowed to Hex Fall in early April, on the basis of corpus quality and per-level win-rate availability from the Paxie CTO.

Experimental Domain

Hex Fall is a triple-match mobile puzzle game. The player sorts coloured slices into matching buckets while contending with ice and pin obstructions, before the board reaches saturation.

Slices: coloured pieces to be cleared.
Bucket: collects one colour.
Ice: frozen, thaws after a fixed number of moves.
Generator: generates buckets.

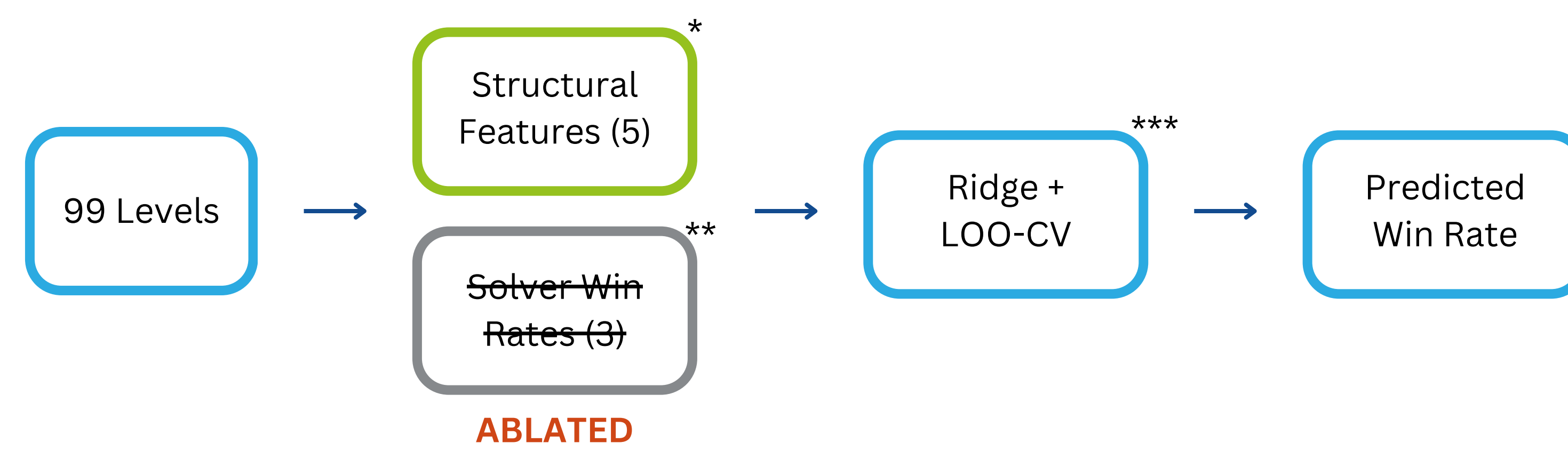
Initial Methodology and Validity Objection

The original plan trained a curiosity-driven PPO agent and used its win rate as the difficulty signal. The advisor raised three objections before any training run:

- RL is disproportionate for a sparse-reward puzzle.
- RL agents converge toward optimal play; humans do not. Agent and human win rate measure structurally different quantities — a validity problem, not a tuning one.
- Reward shaping changes what is optimised, not how well, so human skill tiers cannot be simulated principally.

Due to these validity concerns, the agent-based approach was discontinued, though the underlying simulation framework was retained.

Bounded-Rationality Solver Ensemble



***Structural:** colours · total pieces · pins · ice · board size

****Solvers:** greedy · lookahead · MCTS

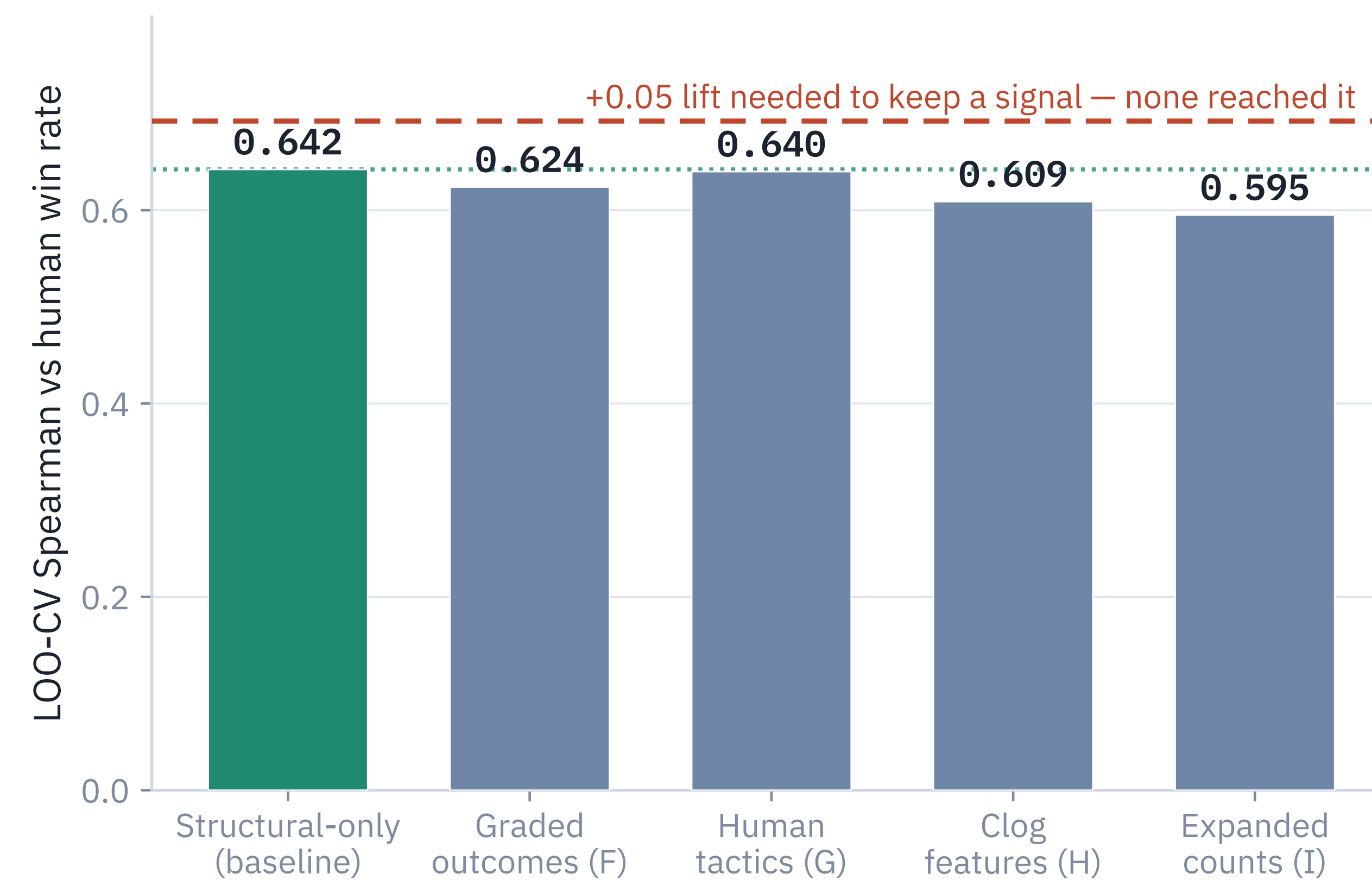
*****Fit:** Ridge regression, leave-one-out cross-validation against 99 real human winrates

Ablation Across Four Signal Axes

Four pre-registered ablations tested whether any behavioural or decision-structure signal would improve on the structural-only baseline. The acceptance threshold was a Spearman lift of +0.05.

- **Graded outcomes (F):** solver partial-progress metrics (fraction cleared, moves survived).
- **Human-tactics player (G):** a hand-built six-component scoring function approximating human heuristics.
- **Clog features (H):** the Unified Clog Hypothesis operationalised as forced-choice count, colour transition density, and hidden-cell ratio.
- **Expanded structural counts (I):** eight additional counts (generators, walls, wood boxes, stack heights, mechanic diversity).

None reached the threshold. The Clog features were verified orthogonal to level size and yet contributed no predictive signal.



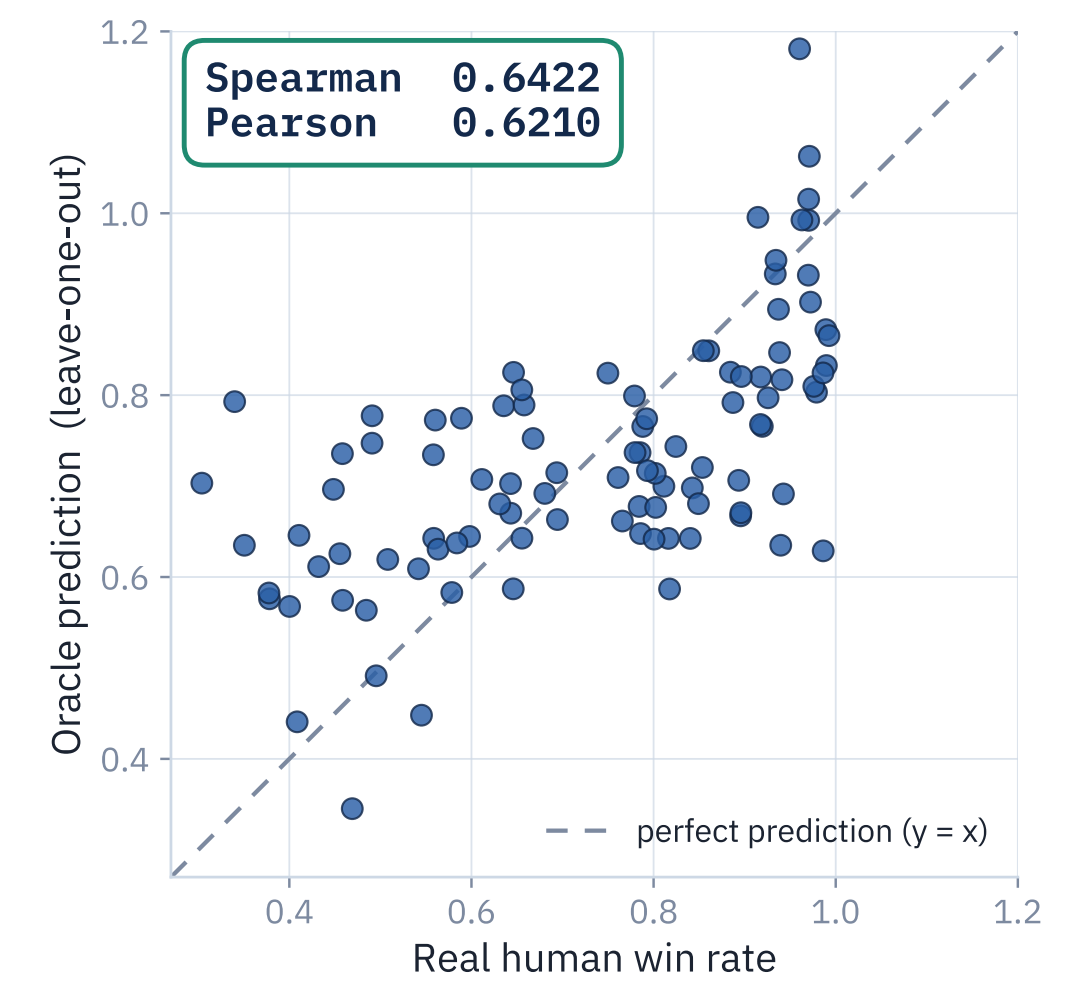
References & Acknowledgements

- Hüseyin Anil Özmen & Paxie Games (Hex Fall corpus and per-level win-rate data)
- Bergdahl, J., Gorrillo, C., Tollmar, K., & Gisslén, L. (2020, August). *Augmenting automated game testing with deep reinforcement learning*.
- Ferdous, Raihana, et al. "Curiosity Driven Multi-agent Reinforcement Learning for 3D Game Testing."
- Hoerl, A. E., & Kennard, R. W. (1970). *Ridge regression: Biased estimation for nonorthogonal problems*.
- Kocsis, L., & Szepesvári, C. (2006, September). *Bandit based monte-carlo planning*.

Empirical Results

Five structural features predict human win rate on the 99-level corpus.

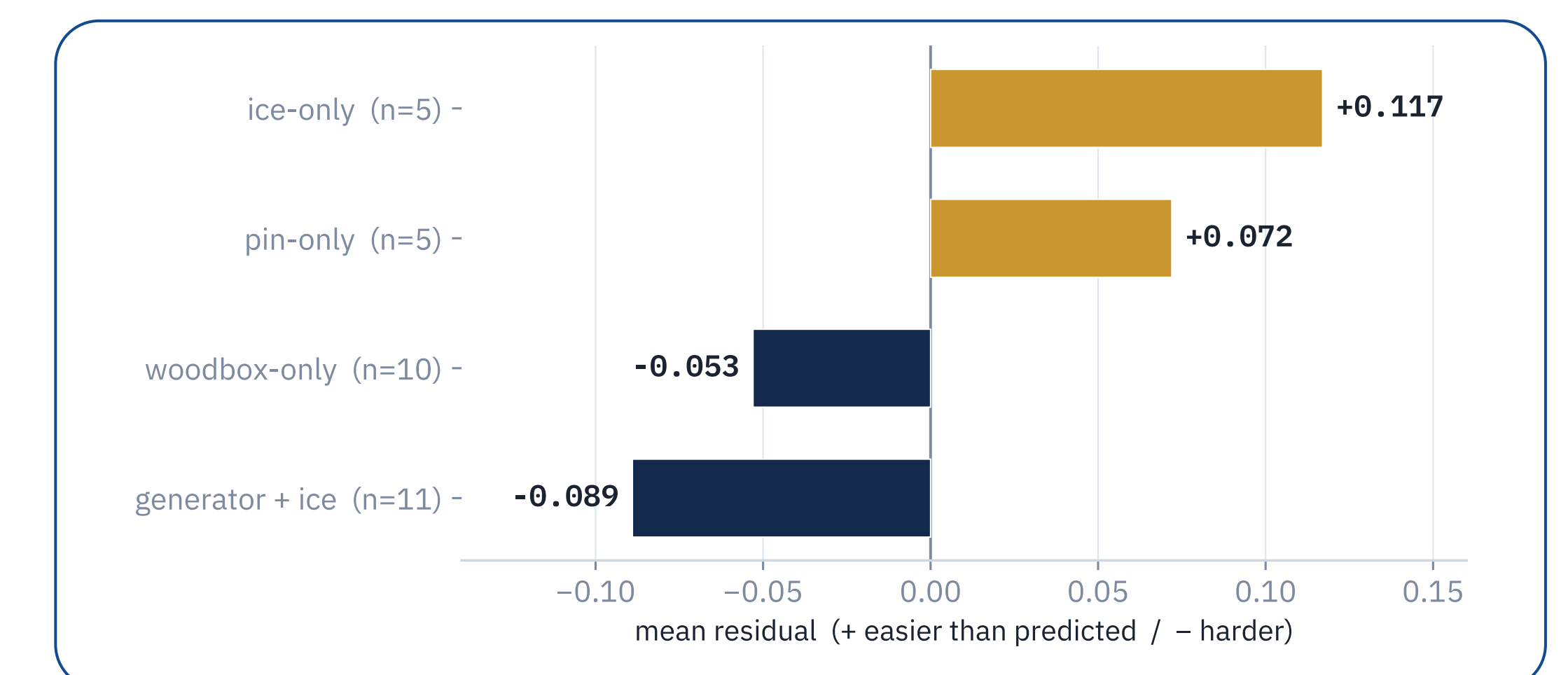
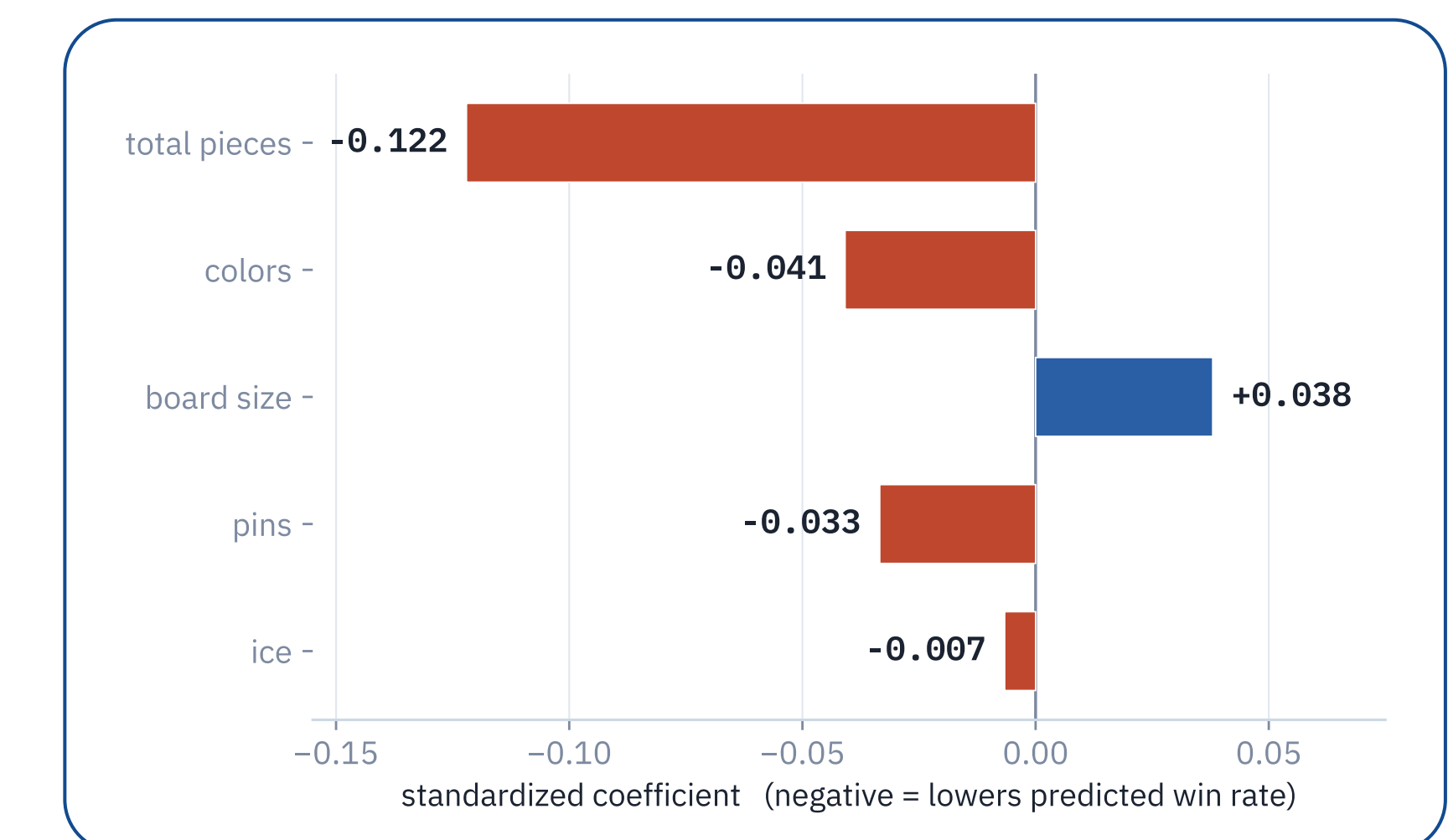
- LOO-CV Spearman 0.6422, Pearson 0.6210, on per-level win rate.
- No behavioural signal, no solver output, no tactical heuristic.
- Predicted versus real win rate, 99 levels (scatter below).



Feature Analysis

Predictive power is concentrated in a single feature, with residuals revealing mechanic-specific blind spots.

- Total slice count carries roughly three times the weight of any other coefficient (left).
- Even features verified orthogonal to slice count contribute no signal in regression. Difficulty on this corpus is dominated by board size.
- Residuals are not uniform across mechanics (right). Ice-only and pin-only levels are systematically easier than predicted; generator-plus-ice combinations are harder than the sum of parts.
- The structural features capture overall difficulty but miss specific mechanic interactions.



Limitations and Future Work

- The result is established on a single game and a single 99-level corpus. Generalisation across titles and corpora is not tested.
- Win rate saturates near 100 percent on easy levels. Average win duration is smoother and predicts at Spearman 0.96 on the same features.
- An open question is whether the structure-dominance finding holds in puzzle genres with richer decision structure, or is specific to Hex Fall.
- A second open question is whether an agent shaped toward characteristically human errors, rather than toward bounded optimality, could recover the predictive signal that bounded-rationality solvers did not.